

E-LEARNING PLATFORM TO HEARING IMPAIRED STUDENTS

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Project Proposal Report

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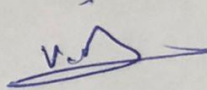
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DECLARATION

DECLARATION

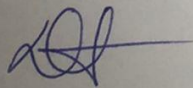
I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution, and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

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ABSTRACT

AI-driven tools and Large Language Models (LLMs) have transformed education into a personalized, adaptive, and widely accessible experience. However, conventional e-learning platforms continue to exclude hearing-impaired learners due to their reliance on auditory content and lack of support for sign language, emotional expression, and multi-sensory communication. This gap is especially pronounced in multilingual and low-resource settings like Sri Lanka, where access to culturally relevant, real-time assistive tools remains limited.

To address this, we propose an innovative AI-powered system: EmotionAware Music — a multimodal, inclusive platform designed to make music education accessible through haptic rhythm, emotion recognition, and Sri Lankan Sign Language (SLSL) animation. The system leverages Music Emotion Recognition (MER) models to classify the valence and arousal of Sinhala and English songs, translating emotional content into dynamic visual and tactile feedback. Crucially, haptic patterns are informed by the UCI Human Activity Recognition Using Smartphones dataset — a rich source of 3-axial accelerometer and gyroscope data sampled at 50Hz from 30 subjects performing daily activities. By mapping natural human movement responses (e.g., walking, sitting) to musical beats and moods, the system generates embodied, intuitive vibrations delivered via smartphones or wearables.

Lyrics are translated into SLSL using sequence-to-sequence NLP models and rendered through a 3D animated avatar built in Blender and displayed in real time using Three.js. A synchronized web interface unifies haptic feedback, SLSL animation, and affective visuals, enabling hearing-impaired students to feel the rhythm, see the lyrics, and experience the emotion of music in a single, cohesive experience.

This system redefines the boundaries of inclusive e-learning by integrating AI, sensor-inspired haptics, and culturally grounded design. It not only enhances accessibility and engagement but also bridges the gap between universal digital education and locally meaningful, domain-specific tools. As demonstrated in this study, such innovations can transform how hearing-impaired students interact with music and multimedia content — paving the way for more inclusive, responsive, and emotionally intelligent learning environments across Sri Lanka and beyond.

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Special thanks go to the Faculty of Computing and the Comprehensive Design and Analysis Project (CDAP) at the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic platform, resources, and research environment necessary to pursue this meaningful study. Their support has been fundamental to our progress and development as researchers.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BPM	Beats Per Minutes
DNN	Deep Neural Network
HAR	Human Activity Recognition
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
LLM	Large Language Model
MFCCs	Mel-Frequency Cepstral Coefficients
SLR52(OpenSLR)	Melody/Tempo Emotional Dynamics in songs (datasets)
ML	Machine Learning
NLP	Natural Language Processing
RNN	Recurrent Neural Network
TPU	Tensor Processing Unit
DEAM	Data set for Emotion Analysis Using Music
API	Application Programming Interface
UI	User Interface
UX	User Experience
HCI	Human-Computer Interaction
F1 Score	Harmonic mean of Precision and Recall
FPS	Frames Per Second

1. INTRODUCTION

1.1 Background of the research

Music is a powerful medium of emotional expression, cultural identity, and social connection. However, for hearing-impaired individuals, the experience of music remains largely inaccessible due to its strong reliance on auditory perception. Unlike traditional audio-based music systems, music accessibility technologies aim to convey not only the rhythmic structure but also the emotional tone and lyrical meaning through alternative sensory channels such as touch (haptics), vision (animation), and movement (embodied feedback). This makes music interpretation particularly valuable for the deaf and hard-of-hearing community, especially in educational settings where emotional and linguistic engagement with music is critical [12] .

Early assistive approaches focused on visualizing sound through spectrograms, flashing lights, or simple beat indicators. However, these methods fail to capture the affective dimensions of music such as valence (positive/negative emotion) and arousal (calm/excited intensity) which are essential for meaningful musical understanding. With the rise of affective computing and multimodal AI, recent systems have begun integrating music emotion recognition (MER) models trained on datasets like DEAM and EMO-MUSIC to classify emotional states from audio features such as tempo, energy, spectral flux, and MFCCs [1]

More advanced developments leverage deep learning architectures including Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), and Transformer-based models—to model complex temporal patterns in music. These models enable end-to-end mapping from raw audio to emotional labels, allowing dynamic adaptation of feedback in real time. Furthermore, self-supervised learning techniques have reduced the need for large annotated datasets, making it feasible to develop systems even in low-resource language settings [13]

Despite these advances, most existing music accessibility tools are designed for Western languages and hearing audiences, with minimal support for local languages like Sinhala or culturally relevant expression. There is currently no publicly available system that translates Sinhala or English songs into a synchronized multisensory experience combining haptic rhythm, Sri Lankan Sign Language (SLSL) animation, and affective visuals. This is a significant gap, given that Sinhala is spoken by over 17 million people, many of whom may benefit from inclusive music education [11]. Moreover, Sinhala music features unique rhythmic structures, tonal variations, and vocal articulations that differ from Western genres. Direct transfer from English-trained models often fails due to linguistic and cultural mismatches,

especially in emotional interpretation and lyrical translation.

Two major challenges hinder the development of inclusive music systems:

Lack of Embodied, Sensor-Informed Haptic Design

While some systems use basic vibration to represent beats, they often lack biological realism and fail to reflect how humans naturally move to music such as nodding, tapping, or swaying. The UCI Human Activity Recognition Using Smartphones dataset offers a solution: it contains 3-axial accelerometer and gyroscope data sampled at 50Hz from 30 subjects performing daily activities (e.g., walking, standing, climbing stairs). By analyzing motion patterns linked to rhythmic movement, this dataset enables the design of naturalistic, embodied haptic feedback that mimics real human responses to music [4]

Absence of Culturally Grounded Visual Feedback

Most systems deliver output only as text or abstract visuals, which are insufficient for deaf users whose primary mode of communication is visual-spatial. There is no existing platform that translates lyrics into Sri Lankan Sign Language (SLSL) using animated avatars. Without accurate, real-time sign language representation, the semantic and poetic content of music remains inaccessible [7]

Another critical gap lies in user feedback design. Current prototypes operate in isolation, delivering fragmented outputs across modalities. A truly inclusive system must synchronize haptics, SLSL animation, and mood visuals in real time, ensuring coherence and reducing cognitive load. This requires precise temporal alignment, emotion-aware rendering, and adaptive feedback based on user interaction. To address these gaps, this project proposes Emotion Aware Music an AI-powered, web-based system that interprets the emotional tone, rhythm, and lyrics of both Sinhala and English songs, and delivers a multisensory experience through:

Haptic feedback informed by human motion data (from the UCI HAR dataset), 3D avatar animation performing SLSL signs synchronized with lyrics, Visual mood indicators reflecting real-time emotional shifts. The system is optimized for real-time performance on standard devices, ensuring accessibility in classrooms, homes, and public spaces. By integrating sensor-inspired haptics, linguistic translation, and affective visualization, this project redefines how music is perceived by hearing-impaired students transforming passive listening into an active, engaging, and emotionally rich experience [3] Ultimately, Emotion Aware Music fills a critical void in inclusive education technology in Sri Lanka, empowering deaf learners to not only "feel the beat" but also understand the message and connect with the emotion of music bridging a long-standing gap between sound, meaning, and inclusion.

1.1. Literature Survey

Music is a universal form of emotional expression, yet it remains largely inaccessible to hearing-impaired individuals due to its reliance on auditory perception. Traditional music systems assume intact hearing, leaving deaf and hard-of-hearing users unable to experience the rhythm, emotion, or lyrical content of songs. To bridge this gap, researchers have explored multimodal music interpretation systems that translate audio into alternative sensory channels primarily visual and tactile (haptic) feedback.

Early assistive approaches focused on sound visualization, using flashing lights, waveforms, or color shifts to represent volume or frequency. However, these methods are limited in conveying emotional tone or rhythmic structure meaningfully. More recent systems leverage affective computing and machine learning to classify music emotions along dimensions of valence (positivity) and arousal (intensity) [1]. Datasets such as DEAM (Dataset for Emotion Analysis using Music) and EMO-MUSIC have enabled the development of Music Emotion Recognition (MER) models using deep learning architectures like CNNs, RNNs, and Transformers. These models extract features such as tempo, beat strength, spectral flux, and MFCCs to predict emotional states in real time [13].

Despite these advances, most existing systems are designed for Western languages and hearing audiences, with minimal support for low-resource languages or culturally relevant expression. There is currently no publicly available system that delivers a synchronized multisensory experience combining haptic rhythm, Sri Lankan Sign Language (SLSL) lyric translation, and affective visuals—especially for Sinhala music, spoken by over 17 million people. This is a critical gap in inclusive education and assistive technology in Sri Lanka.

A key limitation in current haptic-based music systems is the lack of embodied realism. Many use arbitrary vibration patterns that do not reflect how humans naturally move to music such as nodding, tapping, or swaying. The UCI Human Activity Recognition Using Smartphones dataset offers a solution. Collected from 30 volunteers (aged 19–48) performing six daily activities WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS, SITTING, STANDING, LAYING the dataset contains 3-axial accelerometer and gyroscope data sampled at 50Hz using a Samsung Galaxy S II mounted on the waist. The signals were pre-processed into sliding windows of 2.56 seconds with 50% overlap (128 readings/window), and a Butterworth low-pass filter (0.3 Hz cutoff) was used to separate body acceleration from gravity. From each window, a 561-feature vector was extracted using time- and frequency-domain variables.

This dataset has been widely used in human activity classification and sensor-based motion analysis. In this project, it is repurposed to model natural human responses to rhythm, enabling the design of biologically inspired haptic feedback. For example, high-energy beats can trigger vibrations resembling WALKING, while calm passages produce subtle pulses akin to SITTING. This approach ensures that tactile feedback is not arbitrary but embodied and intuitive, enhancing the user's ability to "feel" the music [4].

Another major research area is visual feedback through animated avatars. Systems like Baldi have demonstrated that talking-head avatars can improve comprehension and engagement, particularly among hearing-impaired users. However, these avatars are typically audio-driven and not integrated with real-time emotion or lyric analysis. Moreover, none support Sri Lankan Sign Language (SLSL), which follows a distinct grammatical structure from spoken Sinhala [7]. Without accurate lyric-to-SLSL translation and synchronized 3D animation, the semantic content of music remains inaccessible. Recent advances in sequence-to-sequence models and 3D animation frameworks (e.g., Three.js, Blender) now make it possible to build real-time, web-based SLSL avatars. However, there is no documented effort to apply these technologies to music interpretation in Sinhala. Additionally, no system synchronizes haptic feedback, SLSL animation, and emotion-based visuals in a unified pipeline resulting in fragmented, cognitively demanding experiences [14].

Furthermore, while self-supervised and transfer learning techniques have reduced data requirements in NLP and speech recognition, they remain underutilized in music accessibility for low-resource languages. For Sinhala, there is no labeled dataset linking audio features to emotional labels or sign language outputs, making end-to-end training challenging.

To address these gaps, this project proposes Emotion Aware Music an AI-powered system that: Uses MER models to interpret emotional content from Sinhala/English songs, Leverages the UCI HAR dataset to generate motion-informed haptic feedback, translates lyrics into SLSL using NLP models and animates them via a 3D avatar, Synchronizes haptics, visuals, and signs in real time through a web-based interface [5].

By integrating sensor data, affective AI, and culturally grounded design, this work pioneers a new paradigm in inclusive music education one that empowers hearing-impaired students in Sri Lanka to not only perceive rhythm but also understand lyrics and experience emotion through a truly

multisensory experience.

1.2. Research Gap

While advancements in affective computing and assistive technologies have enabled new ways of experiencing multimedia content, music remains largely inaccessible to hearing-impaired individuals, particularly in multilingual and multicultural contexts such as Sri Lanka. Existing music players and educational platforms predominantly rely on auditory perception, offering minimal or no alternative sensory modalities for emotional and rhythmic interpretation. A critical review of current research and systems reveals several interrelated gaps that this study aims to address.

- **Lack of Emotion-Centric Multisensory Music Systems for the Hearing Impaired**
Most assistive technologies for music accessibility focus solely on visualizing sound through spectrograms or flashing lights, which fail to convey the emotional tone, rhythm, or lyrical meaning of songs. These systems often neglect the affective dimensions of music such as valence, arousal, and mood rendering the experience flat and disconnected from the intended emotional narrative. There is a notable absence of integrated frameworks that translate musical emotion into non-auditory sensory feedback in a meaningful, synchronized manner [3]
- **Absence of Culturally and Linguistically Inclusive Music Interpretation Tools**
Current music emotion recognition (MER) models are primarily trained on Western music using datasets like DEAM or EMOMUSIC, with limited representation of South Asian musical structures, scales (e.g., raga-based compositions), or lyrical content in local languages such as Sinhala. Furthermore, no existing system supports real-time translation of Sinhala or English lyrics into Sri Lankan Sign Language (SLSL) with animated avatars, depriving deaf users of both linguistic and emotional context during music engagement [8].
- **Limited Integration of Haptic Feedback for Rhythmic and Emotional Expression**
Although haptic devices (e.g., smartwatches, wearables) are increasingly used in human-computer interaction, their application in music-based emotion delivery remains underexplored. Most vibration patterns in existing systems are simplistic and asynchronous, failing to map tempo, beat intensity, energy, or spectral flux to dynamic haptic profiles that reflect emotional shifts in music[3]. There is a lack of research on how mobile or wearable vibration motors can simulate not just rhythm, but also emotional texture such as excitement, sadness, or tension through modulated frequency, amplitude, and duration.

- **No End-to-End Synchronized Multimodal Experience** Despite isolated advances in sign language animation, music emotion recognition, and haptic feedback, there is currently no unified system that synchronizes lyrics (via SL animation), rhythm (via haptics), and mood (via visual cues or color dynamics) in real time. This fragmentation limits the coherence and immersion of the music experience for hearing-impaired users. Without temporal alignment across modalities, the cognitive load increases and emotional resonance diminishes.
- **Scarcity of User-Centered Evaluation in Deaf Education Contexts** Few studies conduct empirical evaluations with hearing-impaired users, especially in low-resource or multilingual settings like Sri Lanka. The effectiveness of multisensory music systems in enhancing emotional understanding, engagement, and learning outcomes among deaf students remains largely untested. There is a pressing need for A/B testing and qualitative feedback to validate the usability, accessibility, and pedagogical value of such tools within real classroom environments [9].
- **Underutilization of Real-World Sensor Data for Adaptive Feedback** Interestingly, while rich sensor data exists for human activity recognition (e.g., the UCI Human Activity Recognition dataset using smartphone accelerometers and gyroscopes), these modalities have not been leveraged to enhance haptic feedback personalization in music systems. This project innovatively draws inspiration from such datasets to model movement-response patterns enabling adaptive vibration profiles that align with natural body responses to music (e.g., nodding, dancing) thus increasing the embodied experience of rhythm [3]

	Research 1 [3]	Research 2 [5]	Research 3 [13]	Proposed
Emotion-aware music interpretation using audio features (tempo, energy, beat)	X	X	✓	✓
Real-time haptic feedback synchronized with musical rhythm and mood	✓	X	X	✓
Sinhala/English lyric translation into SLSL with animated avatar	X	✓	X	✓
Multimodal synchronization (haptics + sign language + visuals)	X	X	X	✓

Table 1 : Research gap illustration

1.3 Research Problem

Recent advancements in affective computing and music information retrieval have enabled systems to interpret the emotional content of music using audio features such as tempo, energy, spectral flux, and beat strength. Publicly available datasets like DEAM and EMO-MUSIC have facilitated the development of robust Music Emotion Recognition (MER) models, primarily for English-language and Western musical genres. These models are increasingly integrated into intelligent systems for mood-based playlist generation, mental health applications, and interactive media. However, these technological advances remain largely inaccessible to hearing-impaired individuals, particularly in low-resource, multilingual settings like Sri Lanka [1]

At the time of writing, there exists no dedicated system that translates the emotional and rhythmic essence of music especially Sinhala songs into a multisensory experience combining haptic feedback, sign language animation, and visual mood cues. This absence creates a significant barrier to inclusive music education and emotional engagement for deaf and hard-of-hearing students, who are often excluded from one of the most universal forms of human expression: music. The core issue stems from a critical lack of integrated, culturally adapted assistive tools that go beyond mere sound visualization. While some research has explored haptic representation of beats or simple rhythm patterns, these systems are typically prototype-level, lack emotional context, and do not support real-time synchronization across modalities [3]. Moreover, they are almost exclusively designed for Western languages and musical styles, ignoring regional linguistic diversity and cultural nuances in musical expression. As a result, Sinhala-speaking hearing-impaired users cannot access the lyrical meaning or emotional depth of songs in their native language [8].

Additionally, no existing system translates Sinhala or English lyrics into Sri Lankan Sign Language (SLSL) using animated avatars. Text-based outputs are insufficient for many deaf learners, whose primary mode of communication is visual-spatial. Without accurate, real-time sign language representation, the semantic and poetic content of music remains inaccessible. This is further compounded by the absence of synchronized haptic rhythms that reflect not just timing, but also emotional intensity such as the difference between a soft, melancholic melody and an energetic, joyful beat [7]

Even more concerning is the lack of user-centered design in current solutions. Most MER or assistive music systems are evaluated in controlled lab environments using neurotypical, hearing participants. There is no documented user testing involving hearing-impaired students in Sri Lankan educational settings, nor any effort to assess cognitive load, emotional resonance, or engagement when interacting with multisensory music interfaces. Without such feedback, systems risk being technically sound but practically irrelevant [9]

Furthermore, while datasets like the UCI Human Activity Recognition Using Smartphones provide rich inertial sensor data (accelerometer and gyroscope) at 50Hz, these have not been leveraged to model or enhance haptic feedback patterns in music systems. This represents a missed opportunity: the same sensor modalities used to detect walking, sitting, or dancing can inform how vibrations should be modulated to reflect natural bodily responses to music making the experience more embodied and intuitive. In broader societal terms, this gap reflects a missed opportunity for inclusive technology innovation in Sri Lanka. With increasing smartphone penetration and digital literacy, there is immense potential to deploy accessible AI-driven tools in schools, rehabilitation centers, and public spaces. A hearing-impaired student should not be limited to watching silent music videos or reading subtitles they should be able to feel the beat, see the lyrics in SLSL, and experience the emotion of a song as fully as any hearing peer [6]

Therefore, there is an urgent need for an end-to-end, real-time, emotion-aware music interpretation system tailored for Sinhala-English bilingual contexts and designed with and for hearing-impaired users. Such a system must integrate: Audio feature extraction for emotion and rhythm detection, Haptic feedback mapping to convey beat and mood via wearable or mobile vibration, Lyric translation and SLSL animation using avatars synchronized with song timing, Multimodal synchronization across haptics, visuals, and sign language, And user validation in real-world educational environments. Until such a system is developed, music will remain an exclusionary domain rather than a tool for emotional development, cultural connection, and inclusive learning[11]

2. OBJECTIVES

2.1. Main Objective

To design, develop, and evaluate an AI-driven Emotion Aware Music system that interprets the emotional tone, rhythm, and lyrical content of music both in Sinhala and English and delivers a synchronized, real-time multisensory experience through haptic feedback, Sri Lankan Sign Language (SLSL) animation, and affective visuals. The system aims to enable hearing-impaired students to perceive and emotionally engage with music in an inclusive, culturally relevant, and accessible manner, overcoming the limitations of auditory-only paradigms in music education. By integrating music emotion recognition models, audio-to-haptic mapping algorithms, and animated SLSL lyric translation, this project seeks to transform passive listening into an embodied, interactive experience. A key innovation lies in the use of sensor-inspired haptic patterns informed by human motion data from the UCI Human Activity Recognition Using Smartphones dataset to generate naturalistic, dynamic vibrations that reflect not only beat and tempo but also emotional intensity and rhythmic structure. This ensures that the tactile feedback resonates with the body's innate responses to music, such as nodding or dancing, thereby enhancing immersion and comprehension. Ultimately, the system will be optimized for deployment on standard mobile devices and evaluated through user testing with hearing-impaired students in Sri Lankan educational settings, ensuring practical usability, low latency, and high engagement. This end-to-end solution redefines music accessibility in South Asia by bridging the gap between affective AI, assistive technology, and inclusive pedagogy.

2.2 Sub Objectives

I. Emotion and Rhythm Feature Extraction

- Extract low-level audio features (e.g., tempo, beat, spectral flux, zero-crossing rate, RMS energy) and high-level perceptual descriptors (e.g., valence, arousal, mood) from Sinhala and English songs using pre-trained Music Emotion Recognition (MER) models trained on datasets such as DEAM and EMO-MUSIC.
- Preprocess and segment audio streams in real time to enable continuous emotional and rhythmic analysis suitable for dynamic feedback generation.

II. Haptic Feedback Modeling Using Motion-Inspired Patterns

- Leverage the UCI Human Activity Recognition Using Smartphones dataset containing 3-axial

accelerometer and gyroscope data sampled at 50Hz to model natural human movement responses (e.g., head nodding, foot tapping, dancing) to different musical rhythms and emotions.

- Map extracted audio features (e.g., beat strength, tempo, energy) to modulated vibration patterns (frequency, amplitude, duration) that mimic embodied reactions to music, enhancing the realism and intuitiveness of haptic feedback through mobile devices or wearables.

III. Multilingual Lyric Processing and SLSL Animation

- Develop a lyric processing submodule that performs real-time translation of Sinhala and English lyrics into Sri Lankan Sign Language (SLSL) using sequence-to-sequence models or rule-based linguistic conversion.
- Generate 3D animated avatars capable of expressing SLSL signs with accurate hand gestures, facial expressions, and body posture, synchronized precisely with the song's timing and emotional tone.

IV. Multimodal Synchronization Engine

- Design a synchronization framework that aligns haptic pulses (rhythm), avatar animations (lyrics), and affective visuals (mood indicators such as color gradients or dynamic backgrounds) in real time.
- Ensure temporal coherence across modalities to minimize cognitive load and maximize emotional immersion for the user.

V. Real-Time System Integration and Deployment

- Build a lightweight, cross-platform application optimized for mid-range smartphones and tablets, capable of running the full pipeline audio analysis, haptic control, and SLSL animation with minimal latency.
- Integrate device vibration APIs (Android Vibration Manager or Wear OS) to deliver responsive tactile feedback and ensure compatibility with affordable wearable hardware.

VI. User-Centered Evaluation and Validation

- Conduct empirical user testing with hearing-impaired students in Sri Lankan schools for the deaf to assess emotional comprehension, engagement, and usability.
- Perform A/B testing comparing experiences with and without haptic support, measuring

outcomes such as accuracy of emotion identification, perceived enjoyment, and sense of inclusion.

- Collect qualitative feedback to refine avatar expressiveness, haptic intensity, and system responsiveness.

VII. Scalability and Educational Integration

- Evaluate the potential for extending the system to support additional languages, music genres, and classroom-based interactive learning modules (e.g., rhythm training, emotional literacy).
- Explore integration with digital education platforms used in special needs schools to promote long-term adoption and scalability across Sri Lanka's inclusive education ecosystem.

3. METHODOLOGY

This section outlines the research methodology and the proposed framework for developing the Emotion Aware Music system an AI-powered, multisensory music interpretation platform designed to make music emotionally and linguistically accessible to hearing-impaired students in Sri Lanka. As the final deliverable is a web-based, cross-platform application, software engineering principles such as modularity, usability, performance optimization, and iterative development will be strictly followed. An iterative agile approach will be adopted, where each development cycle includes design, implementation, testing, user feedback, and refinement. This ensures the final system achieves high accuracy, low latency, and intuitive interaction across all sensory modalities

The Emotion Aware Music system will be structured into four core components:

I. Music Emotion & Rhythm Analysis Engine

Captures audio input from uploaded music files or live streams (Sinhala and English songs). Extracts low-level audio features (e.g., tempo, beat, zero-crossing rate, spectral centroid, RMS energy) and mid-level descriptors (e.g., beat strength, harmonic-to-noise ratio) using Librosa and Essentia. Applies pre-trained Music Emotion Recognition (MER) models (trained on datasets like DEAM and EMO-MUSIC) to classify emotional dimensions such as valence (positivity) and arousal (intensity) in real time. Segments audio into fixed windows (e.g., 2–3 seconds) for continuous emotion tracking, enabling dynamic adaptation of feedback across the song [8]

II. Haptic Feedback Modeling Module

Maps extracted audio features to vibration patterns delivered via mobile phone motors or wearable devices (e.g., smartwatches, armbands).

Leverages the UCI Human Activity Recognition Using Smartphones dataset to inform haptic design:

Uses 3-axial accelerometer and gyroscope data sampled at 50Hz to model natural body movements (e.g., head nodding, foot tapping) associated with different rhythms and emotional intensities. Translates motion patterns into modulated haptic profiles varying frequency, amplitude, and duration to simulate embodied musical experiences. Implements adaptive haptic intensity scaling based on genre, tempo, and emotional valence (e.g., stronger pulses for high-energy beats, softer rhythms for calm passages). Ensures compatibility with Android's Vibration API and wearable platforms for real-time tactile delivery

III. Multilingual Lyric Processing & SLSL Animation System

Processes Sinhala and English lyrics in sync with the audio timeline. Translates lyrics into Sri Lankan Sign Language (SLSL) syntax using a hybrid approach combining rule-based linguistic conversion and fine-tuned sequence-to-sequence models (e.g., mBART or custom transformer). Generates 3D animated avatars using morph-target animation (blendshapes) or skeletal rigging to represent SLSL signs, facial expressions, and mouth movements (visemes) [8] Synchronizes avatar gestures precisely with lyric timing, beat, and emotional cues to ensure coherence and reduce cognitive load. Supports customization of avatar appearance, sign speed, and display layout for improved accessibility.

IV. Web-Based Multimodal Interface

Develops a responsive, user-friendly web application using React.js (frontend) and Node.js/Express (backend), optimized for mid-range smartphones and tablets. Integrates real-time audio processing, haptic control (via Bluetooth or device API), and SLSL avatar rendering within a single interface. Displays synchronized outputs: Visual mood indicators (e.g., color gradients, animated backgrounds reflecting valence/arousal),

Haptic intensity meter,

SLSL avatar window,

And optional text subtitles in Sinhala/English.

Includes playback controls, volume-independent sensitivity settings, and user preferences (e.g., haptic intensity, avatar visibility).

Designed with inclusive UX principles, ensuring accessibility for users with diverse communication needs.

Integration & Synchronization Framework

A central synchronization engine ensures temporal alignment across all modalities:

Audio → Haptics: Real-time beat and energy mapping.

Audio → SLSL: Precise lip-sync and gesture timing.

Emotion → Visuals: Dynamic background color shifts (e.g., blue for calm, red for excitement).

Uses Web Audio API and WebSocket communication for low-latency coordination between components.

Validation & Evaluation

Conduct A/B testing with hearing-impaired students in Sri Lankan schools:

Compare comprehension and emotional engagement with and without haptic feedback.

Measure accuracy of emotion identification, user satisfaction, and sense of inclusion.

Collect qualitative feedback through interviews and surveys to refine avatar expressiveness, haptic comfort, and interface usability.

Evaluate system performance using metrics such as:

Latency ($< 200\text{ms}$),

Synchronization error ($\pm 50\text{ms}$),

Emotion classification accuracy,

And user engagement scores.

This methodology ensures the development of a fully functional, real-time, end-to-end Emotion Aware Music system that is not only technically robust but also culturally relevant, linguistically accurate, and deeply inclusive. By leveraging sensor data for haptic realism and integrating SLSL animation with emotional feedback, the project pioneers a new paradigm in accessible music education for the hearing-impaired community in Sri Lanka and beyond

3.1 System Architecture Diagram

The process begins with the User Interface (UI) Module, a web-based dashboard tailored for students and educators. This module captures diverse inputs webcam data (gestures, emotions, lip movements) and lecturer content (text/audio) and allows customization of avatar styles (teacher, realistic, cartoon). It serves as the entry point, delivering real-time outputs such as SLSL videos, emotion feedback, haptic cues, and progress reports, fostering an interactive and adaptive learning experience.

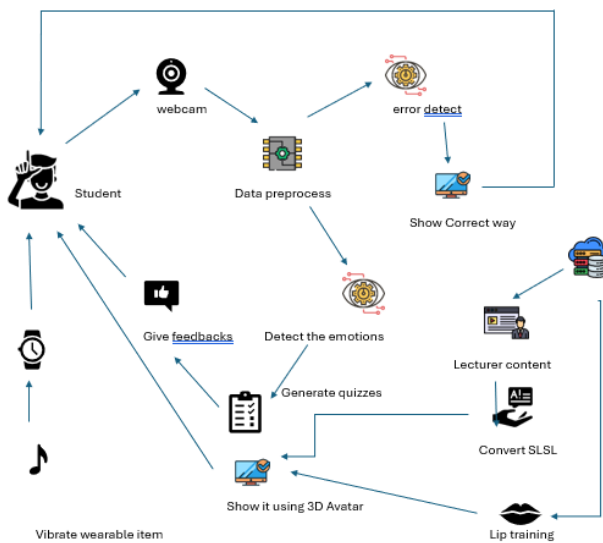
Inputs are processed by the Input Processing Module, which employs Natural Language Processing (NLP) for text/audio and computer vision for gesture and emotion analysis. This module segments data into culturally relevant formats, feeding structured outputs (phonemes, gesture features, emotion

metrics) to subsequent AI-driven components [8]. The AI-Driven Sign Language Generation Module utilizes Generative Adversarial Networks (GANs), Transformers, and Convolutional Neural Networks (CNNs) to convert processed content into SLSL video sequences. It incorporates facial expressions and non-manual markers, enabling bi-directional communication, and links to the Avatar Customization and 3D Animation Module. This module, powered by Blender and Three.js, animates 3D avatars with customizable styles, ensuring natural and culturally resonant interactions [8].

The Emotion Recognition with Real-Time Feedback Module leverages CNN/ResNet and reinforcement learning to analyze facial expressions, gaze, and posture, segmenting lessons into 20-minute stages with adaptive prompts and mini-games based on concentration scores. The Multimodal Feedback Mechanism Module enhances engagement by delivering haptic vibrations and visual aids, while the Lip-Reading and Adaptive Learning Module uses CNN-LSTM and phoneme mapping for lip-reading quizzes and individualized pacing.

Outputs from these modules converge at the Output Rendering Module, which synchronizes SLSL videos, haptic feedback, and visual dashboards with audio/text, adhering to Universal Design for Learning (UDL) principles and adapting to device capabilities. The rendered outputs feed into the System Output, providing adaptive lessons, feedback, and progress reports, with a feedback loop back to the UI for continuous improvement .

The Middleware API Layer, built on Flask/FastAPI, coordinates real-time bi-directional communication (Sinhala-to-SLSL, SLSL-to-Sinhala) across all modules, ensuring low latency and scalability. The Database & Storage Module, implemented with MongoDB, securely stores SLSL datasets, user profiles, emotion logs, and performance data, supporting all components with 99.9% uptime.



3.2 Component specific System Diagram

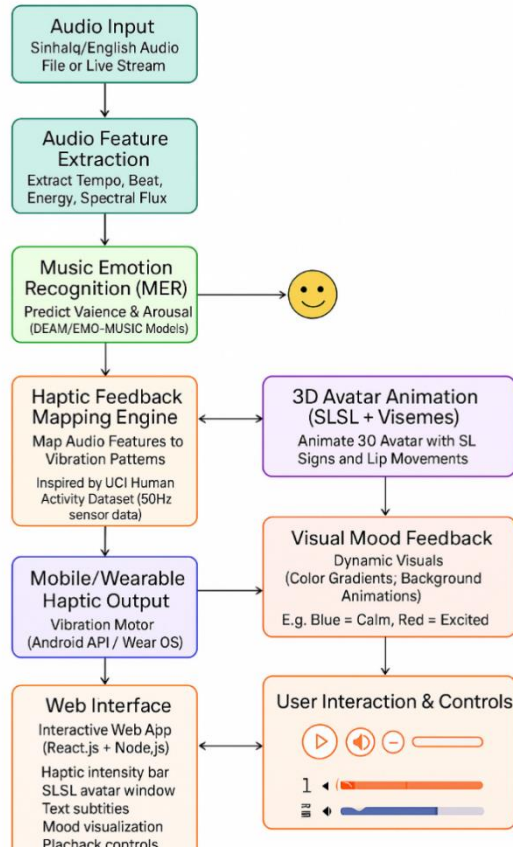


Figure 2: Component specific system diagram

The system chart represents the component diagram that illustrates the internal workflow of the Emotion Aware Music platform an AI-powered, multisensory music interpretation system designed to make music accessible and emotionally meaningful for hearing-impaired users in Sri Lanka. At its core lies a real-time multimodal pipeline that transforms auditory content into synchronized haptic, visual, and linguistic feedback, forming the backbone of an inclusive music experience.

The process begins with the audio input stage, where the system accepts either a live audio stream (e.g., from a microphone) or a pre-uploaded music file in Sinhala or English. This raw audio data is then processed by the Audio Feature Extraction Module, which computes low-level acoustic features such as tempo, beat strength, spectral flux, zero-crossing rate, and RMS energy using tools like Librosa and Essentia. These features serve as the foundation for understanding both the rhythmic structure and emotional tone of the music .

The extracted features are passed to the Music Emotion Recognition (MER) Engine, which leverages pre-trained models trained on datasets such as DEAM and EMO-MUSIC to classify the song's emotional state along dimensions of valence (positive/negative emotion) and arousal (calm/excited intensity). This enables the system to interpret not just what is being played, but how it makes the listener feel

In parallel, the Lyric Processing Module analyzes the song's lyrics, extracting text in real time and translating them into Sri Lankan Sign Language (SLSL) using a hybrid approach combining rule-based grammar rules and fine-tuned sequence-to-sequence neural models. This translation ensures that the semantic content of the music remains accessible to deaf users who rely on visual language.

The Haptic Feedback Modeling Module takes the audio features and emotional classification as inputs and maps them to dynamic vibration patterns delivered through mobile device motors or wearable sensors. Crucially, this module draws inspiration from the UCI Human Activity Recognition Using Smartphones dataset, which contains 3-axial accelerometer and gyroscope data sampled at 50Hz during activities like walking, sitting, and dancing. By analyzing how humans naturally move to music such as nodding heads or tapping feet the system generates realistic, embodied haptic responses that mimic natural bodily reactions. For instance, a high-energy beat triggers stronger, faster vibrations, while a slow, melancholic melody produces gentle, sustained pulses

These haptic patterns are then sent to the Mobile/Wearable Output Layer, which interfaces with the device's vibration API (e.g., Android Vibration Manager) to deliver tactile feedback in real time. This allows users to "feel" the rhythm and emotion of the music through their bodies, enhancing immersion and emotional connection.

Simultaneously, the SLSL translations are fed into the 3D Avatar Animation Engine, which renders a virtual character capable of performing accurate sign language gestures, facial expressions, and lip movements (visemes). The avatar uses morph-target animation (blend shapes) or skeletal rigging to ensure smooth, natural-looking motion, synchronized precisely with the timing of the lyrics and musical beats

To enrich the emotional experience further, the Visual Mood Feedback Module generates dynamic visual cues such as color gradients, animated backgrounds, or abstract motion graphics that reflect the current emotional state of the music. For example, a joyful segment might be accompanied by warm, pulsating

red-orange visuals, while a calm passage displays soft blue tones. All these outputs haptic feedback, SLSL avatar animation, mood visuals, and optional text subtitles are integrated into the Web Interface, a responsive, user-friendly application built with React.js and Node.js. The interface presents all modalities in a unified layout, allowing users to toggle between feedback types and customize settings such as avatar style, haptic intensity, and sign speed [10]. Finally, the User Interaction Layer provides playback controls, volume-independent sensitivity adjustments, and accessibility options, ensuring the system is intuitive and adaptable to individual needs. Real-time synchronization across all components ensures that haptics, visuals, and signs align perfectly with the music, minimizing cognitive load and maximizing comprehension. This coordinated, multimodal feedback system combining emotion-aware audio analysis, sensor-informed haptics, SLSL animation, and affective visuals creates a deeply immersive and inclusive music experience (Paula Andrea Rodríguez-Correa et al., 2023). It empowers hearing-impaired students in Sri Lanka to not only understand the words of a song but also feel its rhythm and grasp its emotional essence, bridging a critical gap in accessible education and cultural participation

3.3 Project Requirements

3.3.1 Functional Requirements

The functional requirements define the core capabilities and behaviors of the Emotion Aware Music system ensuring it effectively translates music into a multisensory experience accessible to hearing-impaired students. These functions are designed to deliver synchronized emotional, tactile, visual, and linguistic feedback in real time

I. Audio Input Processing

Accept audio input from uploaded music files (MP3, WAV) or live microphone streaming in both Sinhala and English.

Support common audio formats and sample rates to ensure broad compatibility.

II. Audio Feature Extraction

Extract time-domain and frequency-domain features such as tempo, beat strength, zero-crossing rate, RMS energy, spectral centroid, and mel-frequency cepstral coefficients (MFCCs).

Segment audio into overlapping windows (2.56 sec, 50% overlap) to enable continuous analysis, inspired by the preprocessing methodology used in the UCI Human Activity Recognition dataset.

III. Music Emotion Recognition (MER)

Classify the emotional state of the music in real time using pre-trained models (e.g., trained on DEAM or EMO-MUSIC) along the dimensions of valence (positivity) and arousal (intensity).

Update emotional predictions dynamically per window to reflect changes in mood across the song

IV. Haptic Feedback Generation

Map extracted audio features to vibration patterns that simulate natural human responses to music (e.g., foot tapping, head nodding).

Leverage motion data from the UCI Human Activity Recognition Using Smartphones dataset recorded at 50Hz via 3-axial accelerometer and gyroscope to model realistic, embodied haptic profiles.

Deliver vibrations through mobile device motors or wearable devices (e.g., smartwatches, armbands) using platform-specific APIs (e.g., Android Vibration Manager)

V. Lyric Translation to SLSL

Process Sinhala and English lyrics in sync with audio timing.

Translate lyrics into Sri Lankan Sign Language (SLSL) syntax using a hybrid system combining rule-based grammar conversion and neural machine translation models.

Ensure culturally appropriate and linguistically accurate sign representation.

VI. 3D Avatar Animation Engine

Animate a virtual character to perform SLSL signs, facial expressions, and lip movements (visemes) using morph-target (blend shape) or skeletal animation techniques.

Synchronize avatar gestures precisely with lyric timing, beat, and emotional shifts to maintain coherence and reduce cognitive load.

VII. Visual Mood Feedback

Generate dynamic visual indicators (e.g., color gradients, animated backgrounds, particle effects) that reflect the current emotional tone of the music.

Example: Warm red-orange hues for high arousal/excitement; cool blue tones for calm/low arousal.

VIII. Web-Based Multimodal Interface

Provide a responsive, user-friendly web application accessible on smartphones, tablets, and desktops.

Display synchronized outputs:

3D SLSL avatar window

Text subtitles in Sinhala/English

Haptic intensity indicator

Visual mood display

Include playback controls (play/pause, seek, volume-independent sensitivity)

IX. Real-Time Synchronization

Ensure temporal alignment between all modalities (audio → haptics → avatar → visuals) with minimal latency.

Maintain synchronization within $\pm 50\text{ms}$ to preserve the natural flow of musical expression

3.3.2 Non-Functional Requirements

These requirements ensure the system is robust, inclusive, efficient, and suitable for real-world deployment in educational and personal settings.

I. Performance

Process audio and generate multisensory feedback with end-to-end latency under 300ms to support real-time interaction.

Optimize haptic and animation modules for smooth performance on mid-range mobile devices.

II. Accuracy

Achieve at least 80% accuracy in music emotion classification (valence/arousal) on test datasets.

Ensure high fidelity in SLSL sign translation and lip-sync accuracy relative to phonetic input.

III. Scalability

Support concurrent usage by multiple users in classroom environments.

Design backend architecture to scale via cloud deployment (e.g., Firebase or AWS) for future expansion.

IV. Compatibility

Operate across modern web browsers (Chrome, Firefox, Safari, Edge) and platforms (Android, iOS, Windows, macOS).

Function effectively with standard smartphone sensors and built-in vibration motors.

V. Usability

Design an intuitive interface with large buttons, clear icons, and customizable settings (e.g., haptic intensity, text size, avatar visibility).

Conduct user testing with hearing-impaired students to refine navigation and accessibility.

VI. Reliability

Maintain consistent performance across varying audio qualities, ambient noise levels, and lighting conditions (for video-based features).

Handle edge cases such as silent segments, overlapping vocals, or fast tempo changes without system failure.

VII. Security

Ensure user privacy by processing audio locally or via encrypted channels.

Do not store personal data or audio recordings without explicit consent.

Comply with ethical standards for assistive technology deployment.

VIII. Maintainability

Implement a modular software architecture allowing independent updates to components:

MER engine

Haptic mapper

SLSL translator

Avatar renderer

Use version control and documentation to support long-term development and academic reuse.

IX. Inclusivity & Cultural Relevance

Prioritize accessibility for users with diverse communication needs, including those who use SLSL as their primary language.

Incorporate Sinhala music styles, lyrical themes, and cultural expressions to ensure emotional resonance.

3.4 Tools and Technologies

3.4.1 Tools

Google Colab

Google Colab is a cloud-based Jupyter notebook environment that allows researchers and developers to write and execute Python code directly in the browser without local setup. It provides free access to high-performance computing resources, including GPUs and TPUs, making it ideal for training and fine-tuning deep learning models used in Music Emotion Recognition (MER) and audio-to-haptic mapping

In this project, Google Colab is used to:

Extract and analyze audio features (e.g., tempo, energy, spectral flux) using libraries like Librosa

and Essentia.

Train and evaluate emotion classification models using datasets such as DEAM and EMO-MUSIC.

Preprocess and model human motion patterns inspired by the UCI Human Activity Recognition Using Smartphones dataset, which contains 3-axial accelerometer and gyroscope data sampled at 50Hz during activities like walking, sitting, and standing. This data informs the design of naturalistic haptic feedback that mimics how people physically respond to music.

Experiment with sequence-to-sequence models for translating lyrics into Sri Lankan Sign Language (SLSL) syntax.

Its seamless integration with TensorFlow, PyTorch, OpenCV, and scikit-learn accelerates prototyping and enables rapid iteration of AI-driven components.

Visual Studio Code (VS Code)

Visual Studio Code is a lightweight yet powerful source-code editor that supports debugging, version control, and extensions for full-stack development. It plays a central role in building the web-based interface and backend services of the Emotion Aware Music application [10]

Key uses include:

Developing the frontend using React.js for responsive UI components, including the SLSL avatar window, haptic intensity meter, and mood visualization panel.

Implementing the backend with Node.js and Express to handle audio processing, model inference, and real-time synchronization

Integrating Three.js for rendering 3D avatars directly in the browser and synchronizing their movements with audio and haptic outputs.

Managing WebSocket communication for low-latency coordination between modules.

Utilizing built-in Git integration and debugging tools to ensure code quality, version control, and collaborative development

VS Code's terminal and extension ecosystem (e.g., Prettier, ESLint, Live Server) streamline the development workflow from model integration to deployment.

Blender

Blender is a free and open-source 3D creation suite used to design, rig, and animate the virtual avatar that performs Sri Lankan Sign Language (SLSL) gestures and lip movements. It supports advanced animation techniques such as morph targets (blendshapes) and skeletal rigging, which are essential for creating realistic and expressive facial animations synchronized with recognized visemes and signs

In this project, Blender is used to:

- Model a customizable 3D avatar with facial and hand articulation capabilities suitable for SLSL expression.
- Create blendshape keys for each viseme and common SLSL handshapes (e.g., fingerspelling, directional verbs).
- Export animated sequences in formats compatible with web rendering engines (e.g., GLTF/GLB for Three.js).
- Ensure cultural appropriateness and clarity in visual communication for Sinhala-speaking deaf users.
- The flexibility of Blender allows for both realistic and stylized designs, enhancing user engagement and comprehension.

Hugging Face

Hugging Face is a leading platform for natural language processing and deep learning model sharing, offering access to thousands of pre-trained models via the Transformers library. While traditionally used for text-based tasks, its models can be adapted for multimodal applications, including lyric translation and emotion-aware processing

In this project, Hugging Face is leveraged to:

- Fine-tune multilingual sequence-to-sequence models (e.g., mBART, T5) for translating Sinhala and English lyrics into SLSL syntactic structure.
- Share and version-control trained models through the Hugging Face Model Hub, enabling collaboration and reproducibility.

- Explore audio-text alignment models that can improve timing accuracy between lyrics and avatar gestures.
- Integrate emotion classification pipelines that enhance the affective understanding of song content.
- By utilizing Hugging Face, the system benefits from state-of-the-art NLP advancements while contributing back to the open research community.

3.4.2 Technologies

UCI Human Activity Recognition Using Smartphones Dataset:

The UCI Human Activity Recognition Using Smartphones dataset provides labeled 3-axial accelerometer and gyroscope data collected at a constant rate of 50Hz from 30 subjects aged 19–48 performing six daily activities: WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS, SITTING, STANDING, and LAYING (Remache-Vinueza et al., 2021). The data was recorded using a Samsung Galaxy S II mounted on the waist, with manual labeling confirmed through video recordings. This dataset is used to model natural human movement responses to rhythm and motion, enabling the system to derive realistic, embodied haptic feedback patterns that mimic how people physically react to music such as head nodding or foot tapping making the tactile experience more intuitive and emotionally aligned

Haptic Feedback Modeling Engine:

A custom haptic engine processes sensor data from the UCI HAR dataset to map audio features (e.g., tempo, beat strength, energy) into dynamic vibration patterns. The sensor signals were pre-processed using noise filters and segmented into fixed-width sliding windows of 2.56 seconds with 50% overlap (128 readings per window). The total acceleration was separated into body motion and gravity components using a Butterworth low-pass filter (0.3 Hz cutoff), allowing precise extraction of motion-related dynamics. From each window, a 561-feature vector was derived using time- and frequency-domain variables, which are used to generate modulated haptic outputs that reflect both rhythm and emotional intensity

Music Emotion Recognition (MER) Models:

MER models analyze audio input to detect emotional content in music along the dimensions of valence (positivity) and arousal (intensity). Using pre-trained models on datasets like DEAM and EMO-MUSIC, and extracting features such as tempo, spectral flux, zero-crossing rate, and RMS energy via Librosa, the system identifies emotional shifts in real time. These predictions drive synchronized visual and haptic feedback, ensuring that the user not only feels the beat but also experiences the mood of the music—enabling a deeper, affective connection for hearing-impaired students.

Sequence-to-Sequence SLSL Translation Module:

This module translates Sinhala and English lyrics into Sri Lankan Sign Language (SLSL) syntax using a sequence-to-sequence neural network trained on linguistic rules and parallel text-sign data. Since SLSL follows a different grammatical structure than spoken language, the model reorders words and inserts appropriate visual-spatial markers to generate accurate sign representations. The output drives the 3D avatar to perform correct hand gestures and facial expressions, ensuring that the meaning of song lyrics is conveyed clearly and culturally appropriately.

Three.js:

Three.js is a JavaScript 3D library used to render the animated SLSL avatar directly in the web application. It supports morph-target animation (blendshapes) for realistic lip movements and facial expressions, synchronized with the translated lyrics and beat timing. The engine ensures smooth playback and real-time responsiveness across devices, allowing the avatar to deliver natural, expressive sign language feedback without requiring high-end hardware.

Python:

Python is the primary programming language used for audio feature extraction, emotion modeling, haptic pattern generation, and backend development. Its simplicity and rich ecosystem libraries such as Librosa, NumPy, Pandas, PyTorch, and scikit-learn make it ideal for building and integrating the AI pipeline. Python is also used to process data from the UCI HAR dataset and interface with web services via Flask, enabling seamless coordination between the frontend

and backend components of the system.

3.5 Implementation

The completed product of the proposed solution is a web-based interactive application designed to interpret the emotional tone, rhythm, and lyrical content of music both in Sinhala and English and deliver a synchronized, real-time multisensory experience through haptic feedback, 3D Sri Lankan Sign Language (SLSL) avatar animation, and affective visuals. The application will act as an inclusive music companion for hearing-impaired students and users in sound-sensitive environments, enabling them to feel the beat, see the lyrics, and experience the emotion of a song through a combination of touch, vision, and movement.

Front End Development –

React.js will be used to develop the front-end interface of the Emotion Aware Music application, providing a responsive, dynamic, and user-friendly experience. The component-based architecture of React enables modular implementation of key UI elements such as the audio input panel, SLSL avatar window, haptic intensity display, mood visualization zone, and subtitle box. With efficient state management and the virtual DOM, React ensures smooth, real-time updates with minimal latency synchronizing haptic pulses, avatar gestures, and visual mood changes precisely with the music timeline. The interface will be optimized for mobile and desktop browsers, ensuring accessibility across a wide range of devices commonly used in Sri Lankan educational settings

Back End Development –

The backend serves as the processing core of the system, responsible for audio analysis, emotion recognition, haptic mapping, and sign language translation. It will be built using Flask (Python), chosen for its lightweight nature, ease of integration with AI/ML models, and compatibility with real-time web applications. Flask will function as the API layer that:

Receives audio input (uploaded or streamed), Extracts audio features (e.g., tempo, energy, spectral flux) using Librosa, Runs Music Emotion Recognition (MER) models to classify valence and arousal, Triggers haptic pattern generation based on sensor-inspired models derived

from the UCI Human Activity Recognition Using Smartphones dataset, And sends synchronized outputs (emotion labels, haptic commands, SLSL sequences) to the frontend

The backend will also interface with PyTorch and TensorFlow for model inference, ensuring low-latency processing suitable for real-time music interpretation.

Implementation of 3D Avatar Animation –

Three.js will serve as the 3D rendering engine to display the SLSL-signing avatar directly in the web browser. The avatar will be animated using morph-target (blendshape) animation to accurately represent hand gestures, facial expressions, and lip movements (visemes) corresponding to the translated lyrics. The viseme and gesture sequences generated by the backend will drive real-time facial and body animations, ensuring natural, expressive, and culturally appropriate signing . Avatars will be modeled and refined in Blender, optimized for performance and lightweight rendering to ensure smooth playback even on mid-range devices.

Database Management –

A MongoDB database will store application metadata, user profiles, system preferences (e.g., haptic intensity, avatar style, language settings), and anonymized usage logs. This data will be used to improve the accuracy of emotion classification models, refine SLSL translation rules, and personalize haptic feedback over time. The schema-less nature of MongoDB allows flexible storage of diverse data types including animation parameters, model versions, and sensor-mapping profiles without disrupting system functionality. All user data will be stored securely and anonymously in compliance with ethical and privacy standards for assistive technology deployment .

3.6 Deployment

Hugging Face Spaces will be utilized to deploy the Emotion Aware Music application, providing a seamless platform for integrating and delivering AI-powered, multisensory music experiences. This environment enables the deployment of trained Music Emotion Recognition (MER) models and SLSL translation systems within an interactive web interface that unifies backend inference with frontend presentation.

Deployment will consist of the following:

- **Model Hosting:** The trained MER models, haptic mapping logic, and SLSL translation pipeline will be hosted on Hugging Face Spaces, leveraging its support for GPU-accelerated inference and containerized environments. This allows real-time processing of audio features and emotion classification while enabling open access for research collaboration and reproducibility
- **Web Hosting:** The React.js frontend and Flask backend will be containerized using Docker and deployed alongside the AI models to ensure consistent performance across devices and sessions. This setup guarantees that the synchronized output haptics, avatar animation, and mood visuals remains stable and responsive regardless of user location or device type
- **Scalability & Access:** The system is designed to support concurrent users in classroom or public settings without performance degradation. It will be accessible through both desktop and mobile browsers, ensuring inclusivity for hearing-impaired students, educators, and Sinhala/English music listeners in sound-sensitive environments

This deployment strategy ensures the application is reliable, maintainable, and user-accessible, making it easy to update with new models or features while offering a smooth, real-time experience for target users.

3.7. Evaluation of the System

The Emotion Aware Music system will be evaluated using a combination of quantitative performance metrics and qualitative user feedback to assess both technical accuracy and real-world usability.

Precision & Recall (Emotion Recognition):

Precision will measure the percentage of correctly predicted emotional states (valence and arousal) out of all predictions made by the MER model. Recall will assess how many of the actual emotional transitions in the music were successfully detected. Given the complexity of emotional perception in multilingual (Sinhala/English) music, a target range of 70–80% for both metrics is expected, balancing sensitivity and specificity in low-resource, culturally diverse contexts

F1 Score:

The F1 Score, as the harmonic mean of precision and recall, will be used to evaluate the overall effectiveness of emotion classification. A target of 75% or higher is set to ensure reliable interpretation of musical mood, forming the foundation for synchronized haptic and visual feedback.

Haptic Synchronization Accuracy:

A specialized metric will evaluate the alignment between music beats and delivered haptic pulses, informed by motion patterns derived from the UCI Human Activity Recognition Using Smartphones dataset. The system will be assessed on its ability to generate vibration sequences that match the natural rhythm of human movement responses (e.g., foot tapping, head nodding). A synchronization accuracy of 85% or higher (within ± 50 ms of beat onset) is targeted to ensure the tactile experience feels intuitive and embodied.

Avatar-Text-Emotion Coherence Score:

This metric evaluates the temporal and semantic alignment between the SLSL avatar gestures, text subtitles, and affective visuals (e.g., color shifts). It ensures that all modalities reflect the same emotional and linguistic content at the correct time. High coherence reduces cognitive load and enhances comprehension, especially for deaf users

User Satisfaction:

Feedback will be collected through surveys and usability testing with hearing-impaired students, special education teachers, and general Sinhala-speaking users. Participants will rate the system on:

Ease of use

Emotional engagement

Clarity of SLSL animation

Helpfulness of haptic feedback

Overall sense of inclusion in music experience

Testing will include A/B comparisons (with and without haptics) to measure the impact of multisensory integration on comprehension and enjoyment.

4. WORK BREAKDOWN STRUCTURE

The following is the finalized WBS for the year, developed after extensive brainstorming on the tasks necessary to complete the research. This WBS outlines the key deliverables and associated tasks, breaking down the project into manageable components. By organizing the work into hierarchical

levels, it ensures that all critical activities are identified and planned for, facilitating resource allocation, progress tracking, and effective project management. This structured approach provides a clear roadmap for the research, guiding the team toward the successful achievement of the project's goals

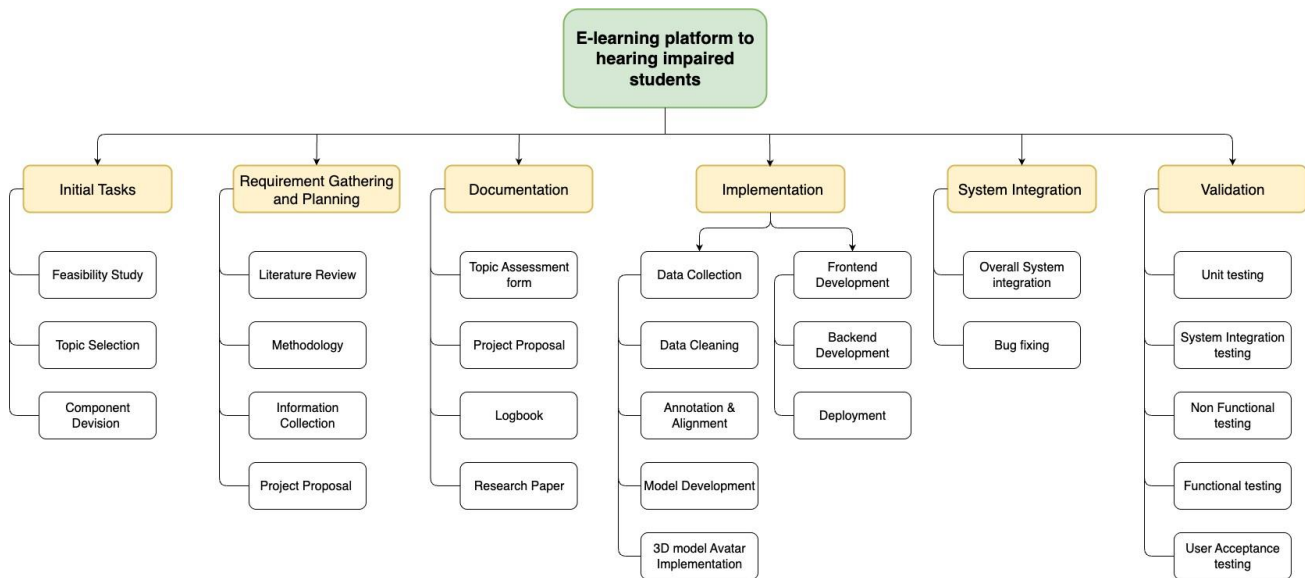
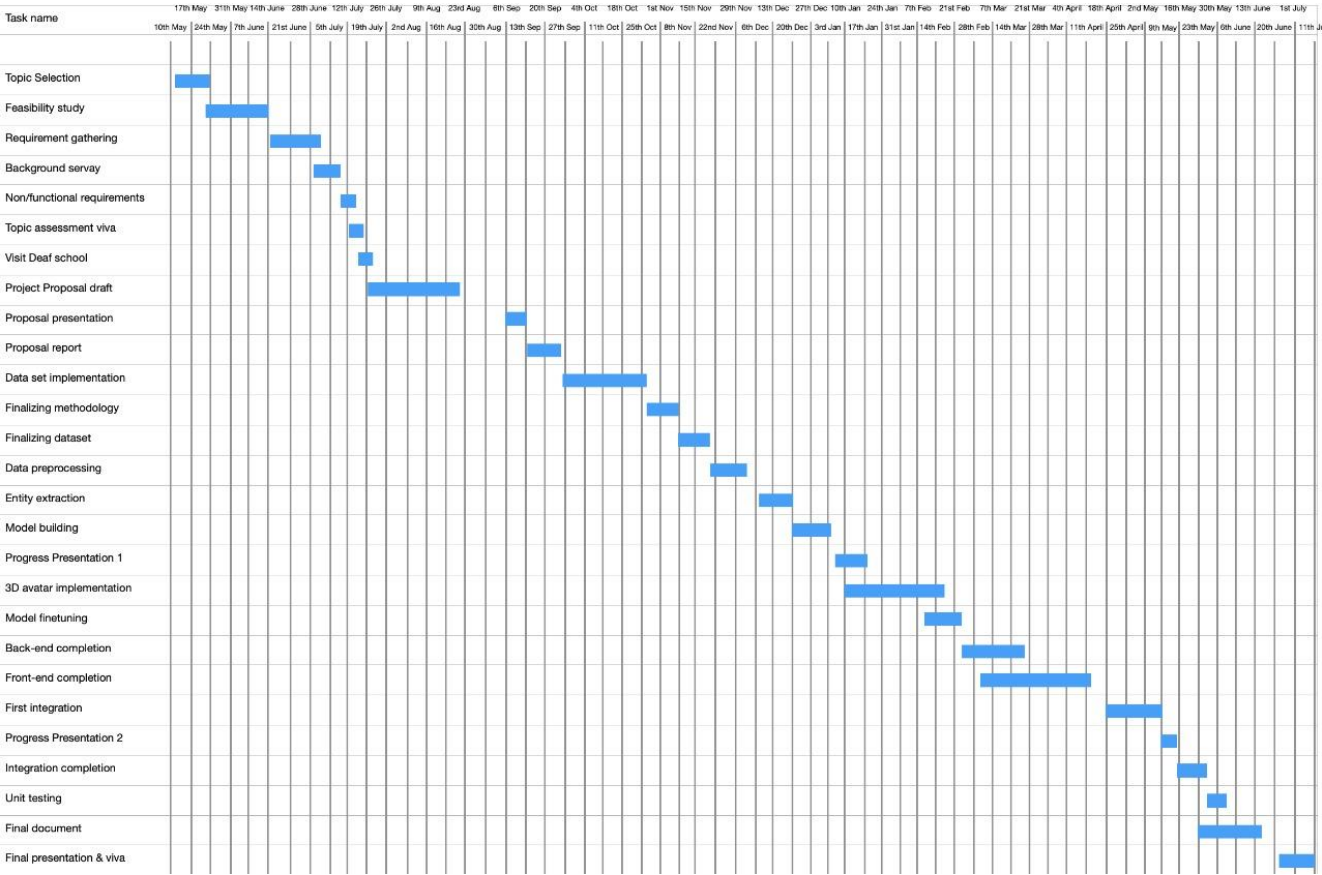


Figure 3: Work breakdown structure

5. GANTT CHART



6. COMMERCIALIZATION OF THE PRODUCT



Figure 4: System Logo

7. BUDGET AND BUDGET JUSTIFICATION

Category	Estimated Cost (LKR)
Dataset Preparation & Annotation	25,000
Internet cost	10,000
Cloud Computing & Hosting	20,000
Development Tools	15,000
Database & Storage	15,000
Testing & Evaluation Tools	10,000
Miscellaneous & Contingency	15,000

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9. APPENDIX

The screenshot displays the Feedback Studio interface. The main document area shows the title "E-LEARNING PLATFORM TO HEARING IMPAIRED STUDENTS" followed by the identifier "25J-26-103", the subtitle "Project Proposal Report", and the author "Viduranga W.R.A.V". The sidebar on the right, titled "Match Overview", shows a similarity score of 8%. Below this, a list of sources is provided with their respective similarity percentages:

Rank	Source	Similarity
1	docslib.org Internet Source	1%
2	ijarose.isroset.org Internet Source	<1%
3	studentsrepo.um.edu... Internet Source	<1%
4	San-Segundo, Rubén, J... Publication	<1%
5	Submitted to Sri Lanka ... Student Paper	<1%
6	Submitted to University... Student Paper	<1%
7	anov.org Internet Source	<1%
8	Submitted to Leiden Un... Student Paper	<1%
9	stackshare.io Internet Source	<1%

The bottom status bar indicates "Page: 1 of 41", "Word Count: 9827", and options for "Text-Only Report" and "High Resolution".